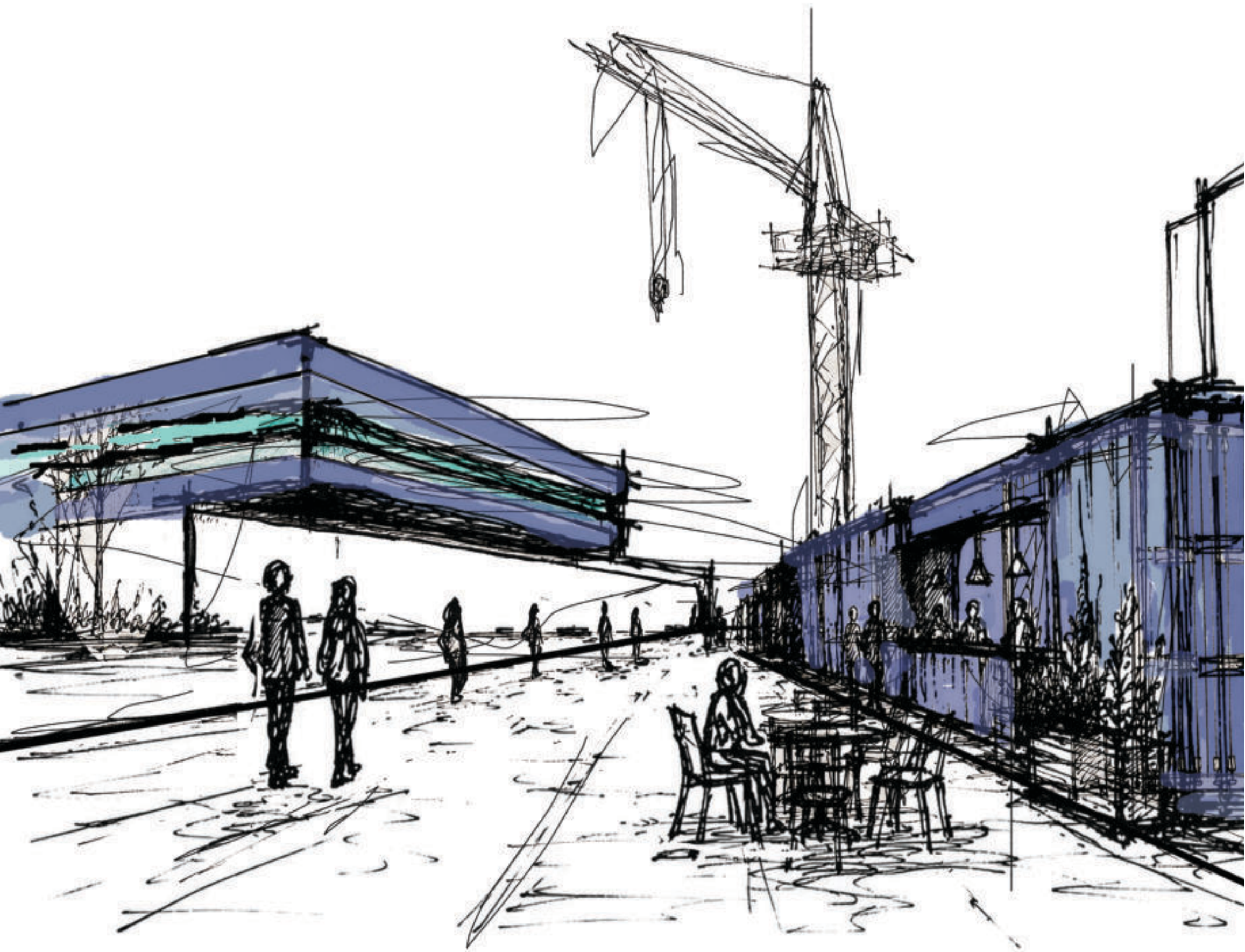


Wind Rós

Sara Sulollari



Rethinking Logistics
AD4

BRIEF

The studio focuses on how industrial and logistics architecture can be reused and transformed into spaces that create more environmental and public value within the city. Instead of seeing logistics buildings only as technical infrastructure, the studio explores how they can become more connected to the city and everyday public life.

Through environmental analysis, research, and design experimentation, the studio investigates topics such as climate, wind, accessibility, and adaptive reuse. The work combines research and design methods to develop more flexible and responsive industrial environments.

Working together with industry partners such as Ramboll, Henning Larsen, and Panattoni, the studio also works with real-world challenges related to the design, performance, and future of logistics facilities.

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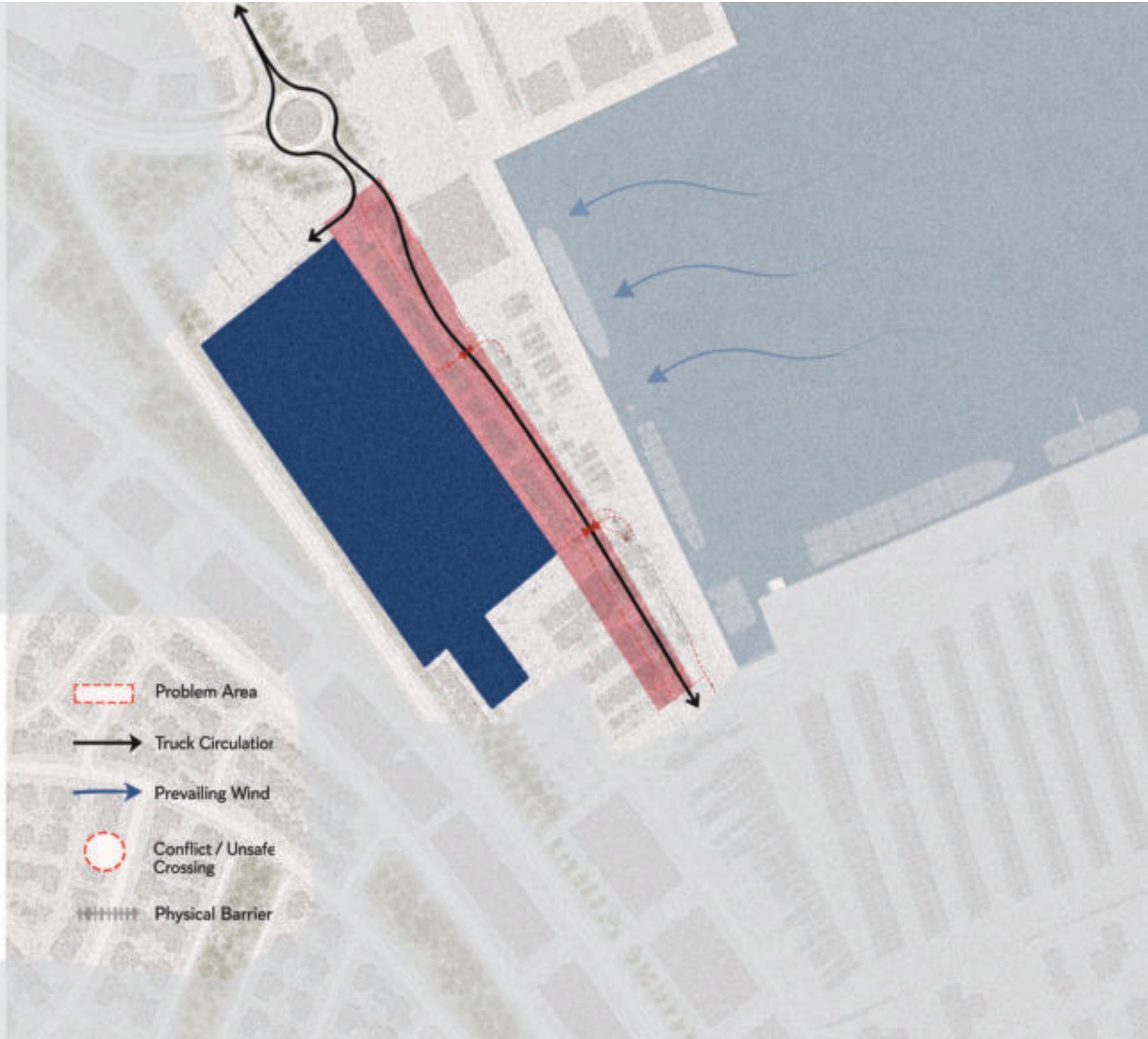
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Wind Rós: INTRO



The Eimskip site at Sundabakki 2 is located along the waterfront in Reykjavik. It is part of the city's main harbor area and is used for shipping, storage, and transport of goods. The site includes large warehouses, open storage spaces, and road connections that support daily port activities. The area is mainly industrial and is designed for efficiency and movement of cargo. Because of its scale and function, it creates a clear separation between the harbor and the nearby residential neighborhoods. Public access to the waterfront is limited, and the area feels closed and disconnected from the rest of the city.

However, the location has strong potential for future development. Its position by the sea and close to the city center makes it an important site for change. There is an opportunity to improve connections between the harbor and the urban area, introduce more public spaces, and create a better balance between industry and everyday city life. With thoughtful design, Sundabakki 2 could become a more open, accessible, and sustainable waterfront area while still keeping its role in logistics and transport.



This diagram shows the main challenges at the Eimskip site in Reykjavik. The red zone highlights the problem area along the waterfront, where most movement and conflicts occur. Truck traffic runs along the site edge, creating a busy corridor with unsafe crossings. The area is designed mainly for industrial use, limiting pedestrian access.

Strong sea winds affect comfort, while physical barriers and large buildings disconnect the site from the surrounding city. Overall, the area feels closed, unsafe, and poorly integrated.

LOCATION// PROBLEM AREA

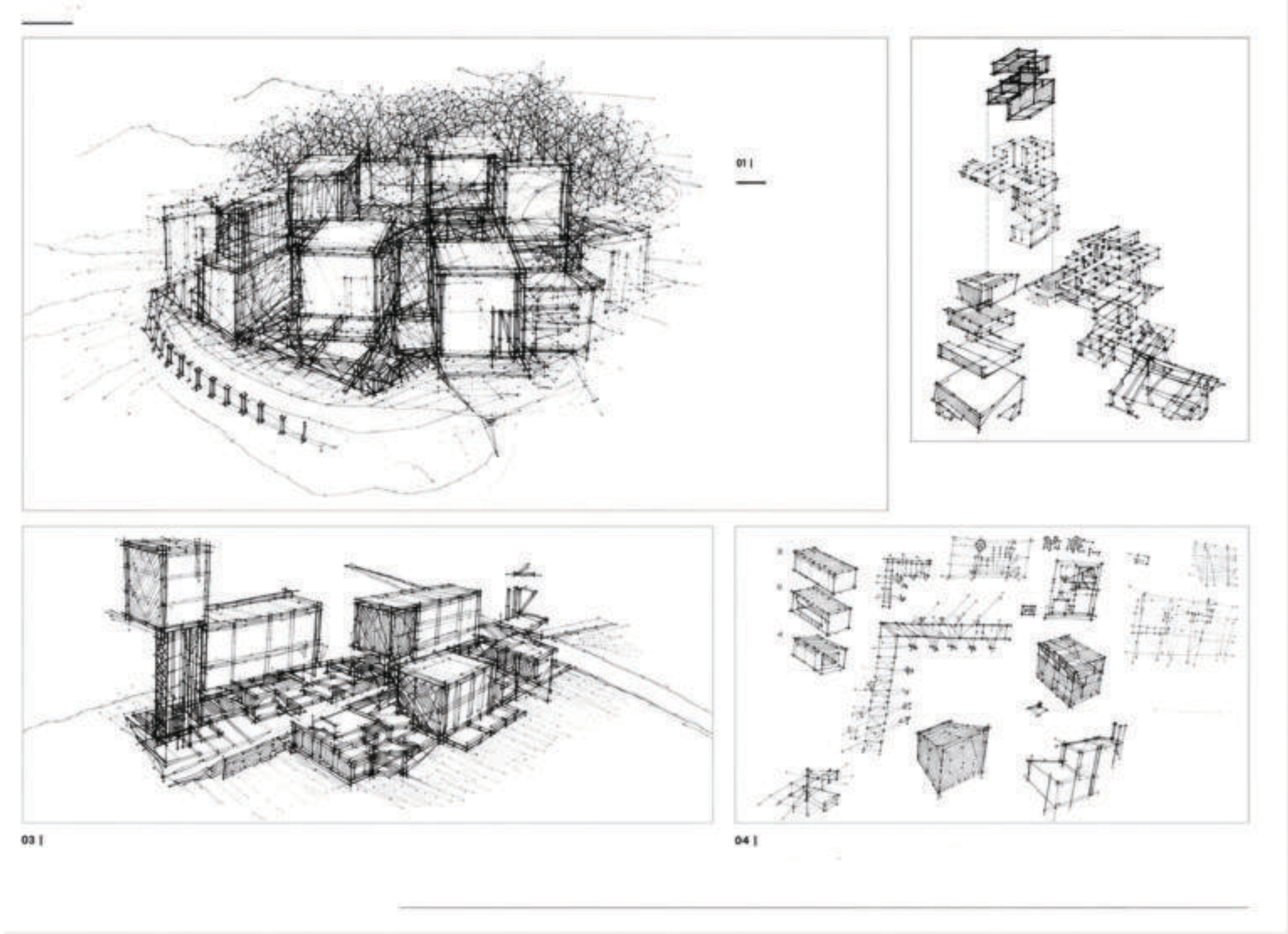
Wind Rós : Pictures



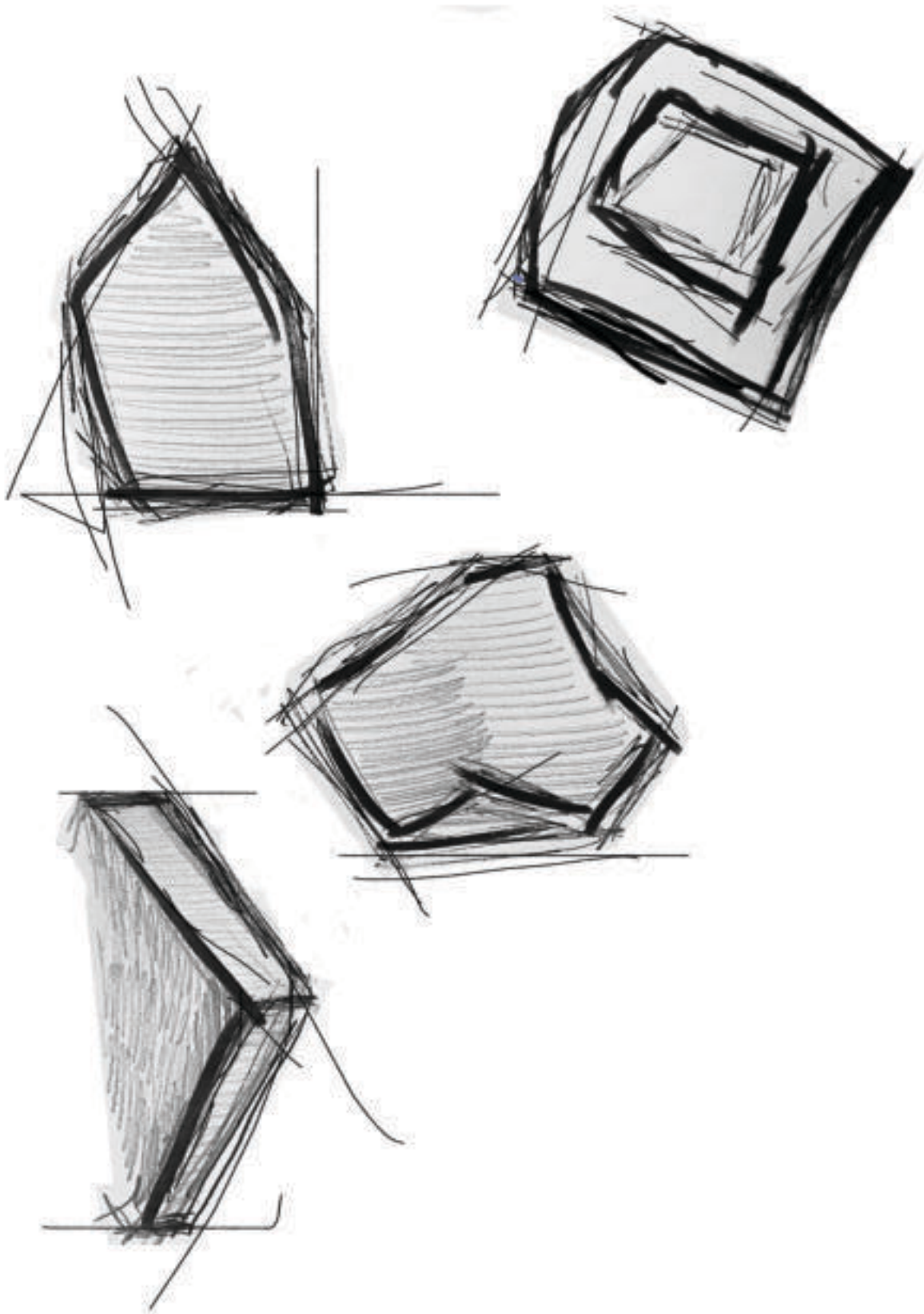
Visual exploration of scale, movement, and infrastructure at the Eimskip harbor in Reykjavík.



Eimskip is one of Iceland's most influential companies, founded in 1914 by around 15,000 shareholders almost 17% of the nation. Its first ships arrived in 1915, opening vital trade routes between Iceland and the rest of the world. Eimskip was a pioneer in modern shipping, introducing Iceland's first refrigerated cargo vessel, which made it possible to export fresh products internationally. In the 1980s, it transformed its operations by adopting container shipping, allowing faster and more efficient global transport. The company later expanded worldwide, opening offices across Europe and North America and growing through key acquisitions. In recent years, Eimskip has focused on innovation and sustainability, investing in electric-powered port equipment and modern vessels to reduce environmental impact. Listed on Nasdaq Iceland, it continues to play a major role in connecting the North Atlantic region to global markets.



Early layout studies testing container organization at Eimskip. The sketches explore spatial efficiency, circulation, and different configuration strategies through quick iteration.



Early sketches exploring massing options for the extension building. Different forms are tested to study spatial relationships and potential integration with the existing structure.

Wind Rós



PK – Turninn, Pálmar Kristmundsson

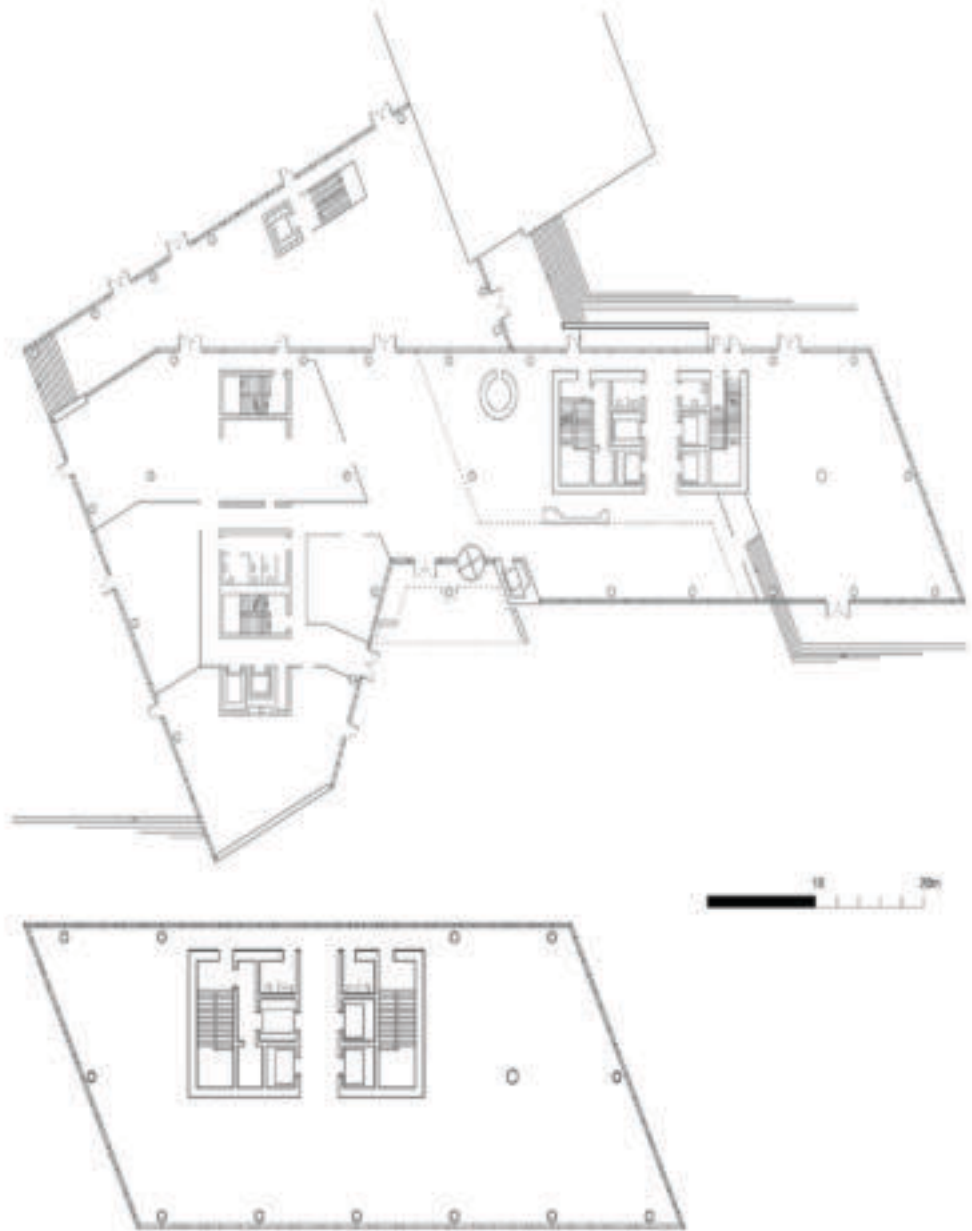
This project became a key source of inspiration, particularly through its response to wind conditions in Iceland. I had the opportunity to speak with architect Pálmar Kristmundsson (Studio PK), who highlighted that, despite the country's extreme climate, Iceland currently has no wind-related building regulations. This often results in outdoor environments that are uncomfortable and poorly protected at the human scale.

Kristmundsson's approach, including wind tunnel testing for the project Turninn, demonstrates the importance of addressing wind as a fundamental design parameter. The harsh conditions surrounding the building further emphasize the need for this approach.

This insight directly informed my project, leading to a strategy that mitigates wind through spatial configuration and the use of reused ship-ping containers as protective elements.

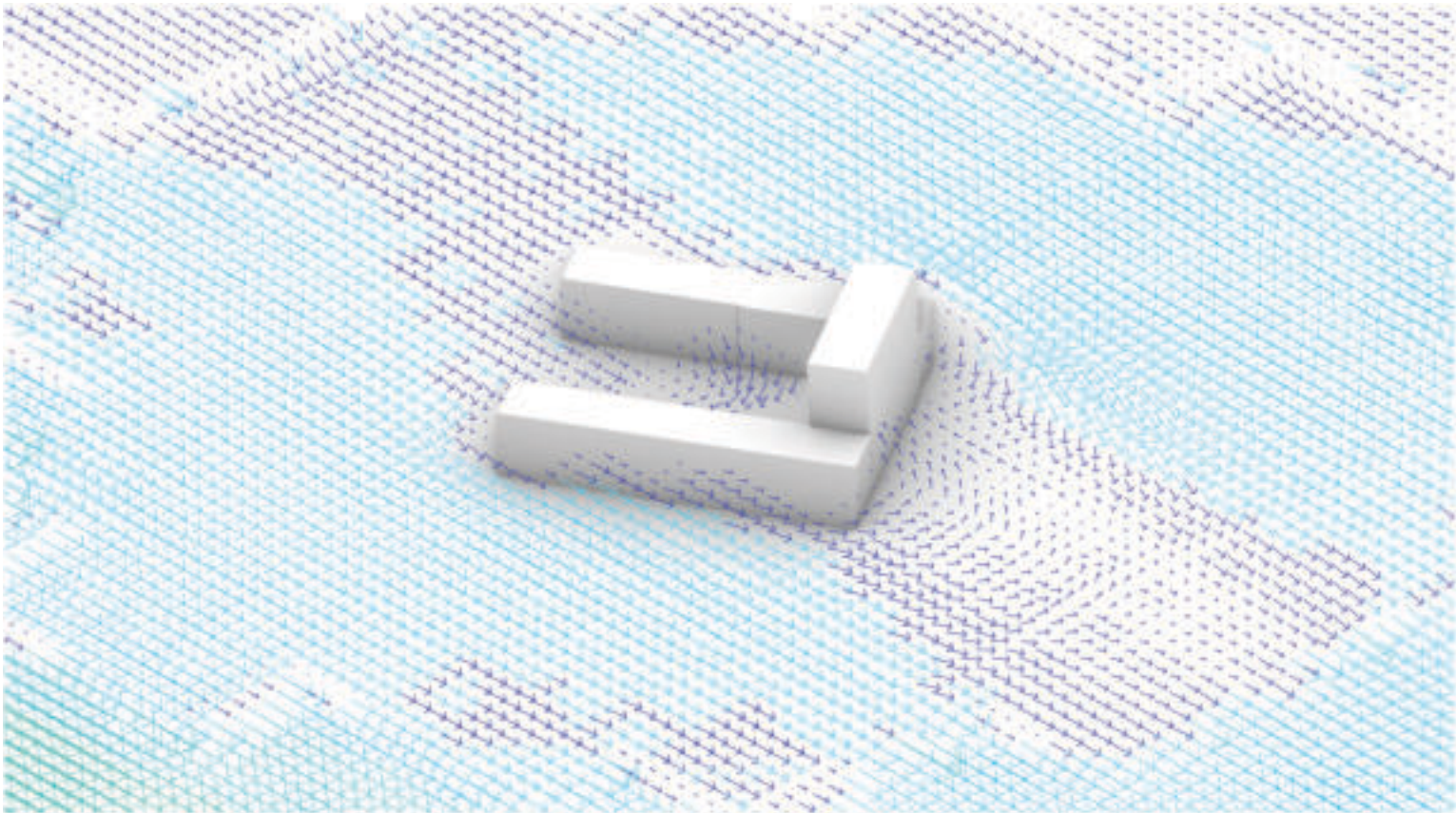
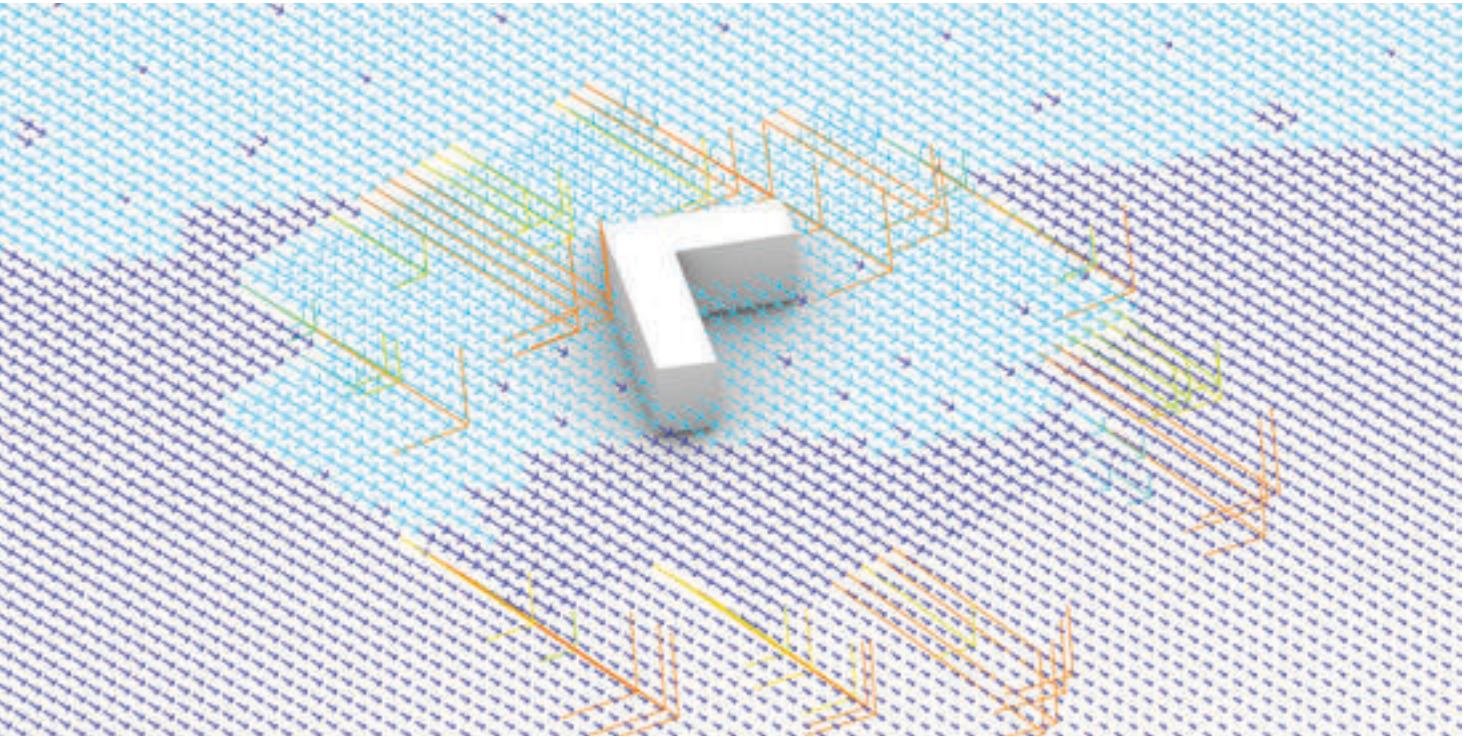
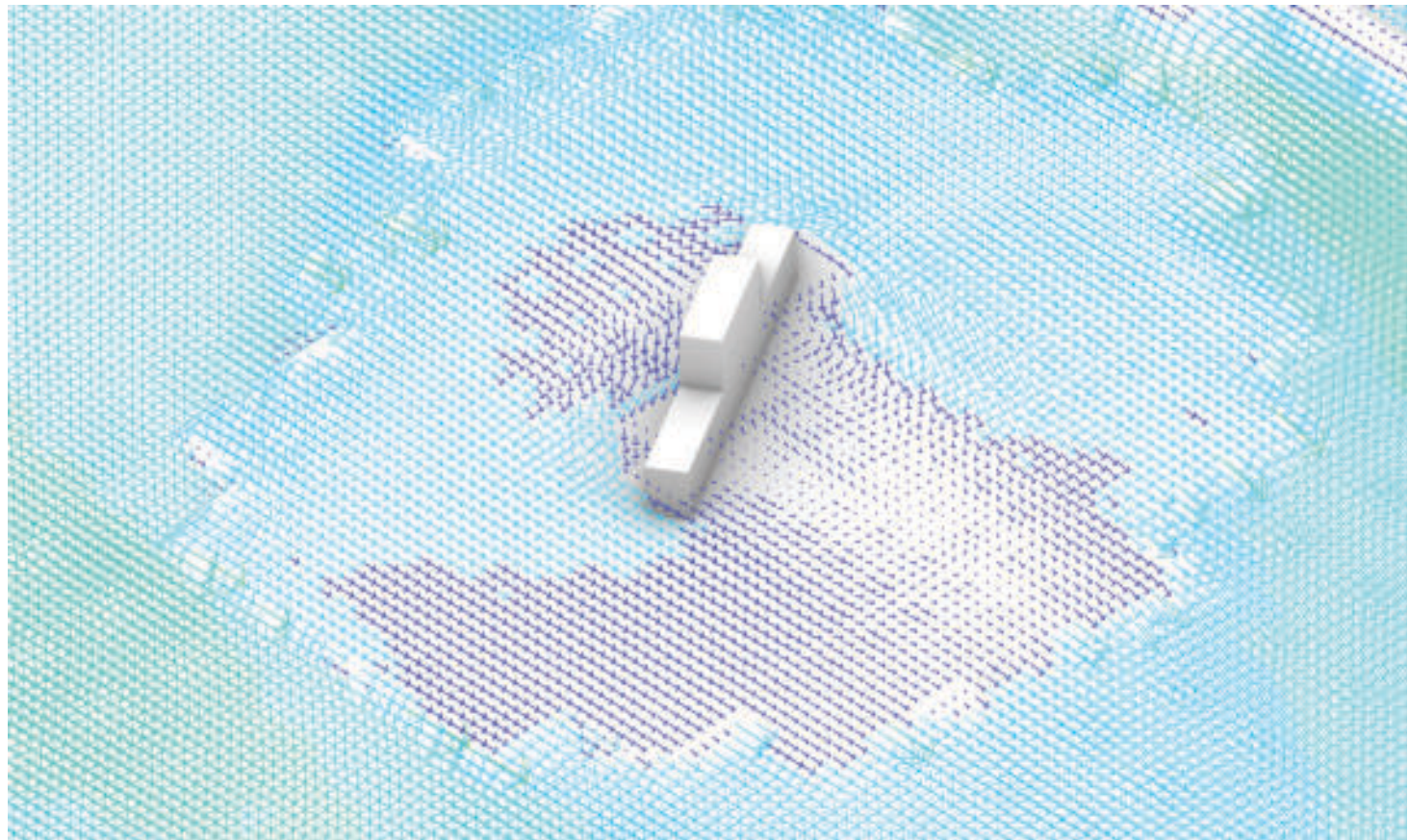


PK – Turninn, Pálmar Kristmundsson



Inspiration I // Inspiration II

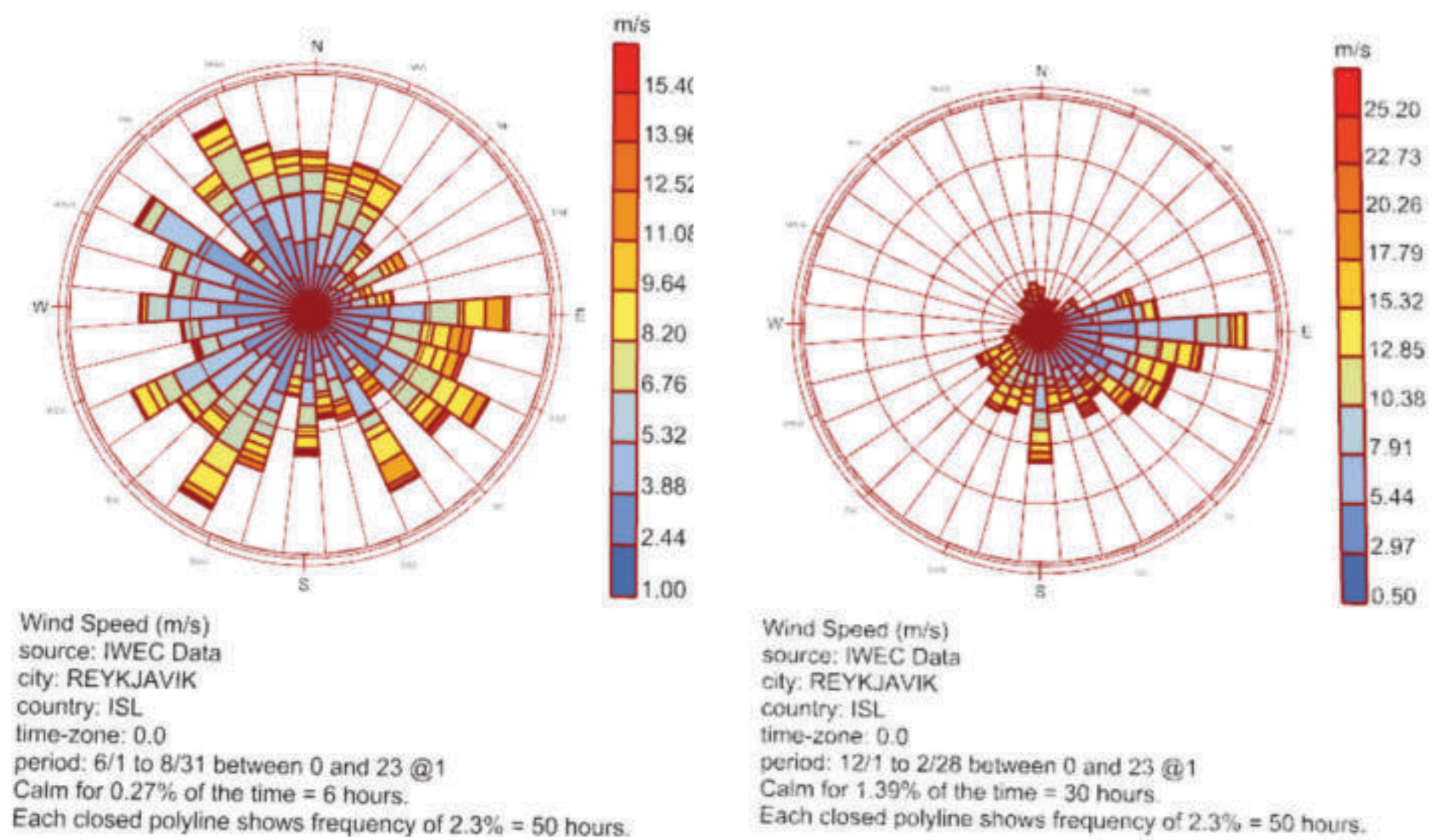
Wind Rós



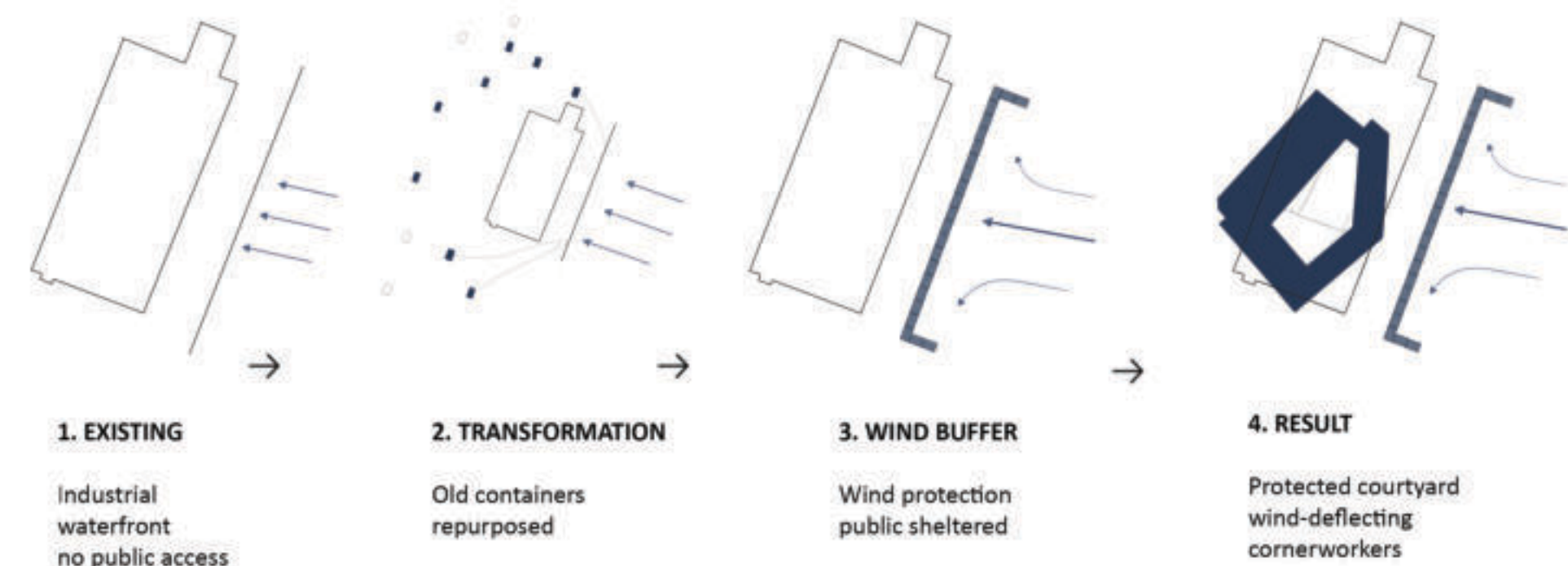
Wind performance was studied using Eddy in Grasshopper to evaluate how different container layouts influence air-flow and create shelter on site. The aim was to identify a configuration that could function effectively as a wind buffer while maintaining spatial efficiency. Three primary arrangements were tested: a linear layout, a corner condition, and a U-shaped configuration. The linear setup offered minimal resistance to wind, allowing airflow to pass through the site with little reduction in speed. The corner condition was explored as a way to deflect and break the wind. Importantly, the angle was intentionally designed to be softer than 90 degrees. Sharp 90-degree corners tend to accelerate wind due to pressure differences and turbulence, which can increase wind intensity at the edges. By opening the angle, this effect was reduced, resulting in a more controlled deflection and a noticeable decrease in wind speed. The U-shaped configuration proved to be the most effective strategy. By partially enclosing space, it was able to both block and redirect airflow, creating a more protected zone within and behind the structure. Simulation results showed that this arrangement reduced wind speeds by approximately 50%, making it significantly more efficient than the other tested options. Based on these findings, the U-shaped layout was selected as the final design strategy for the container wall, functioning as a wind buffer that improves environmental comfort and site usability.

Wind Simulation I // Wind Simulation II

Wind Rós



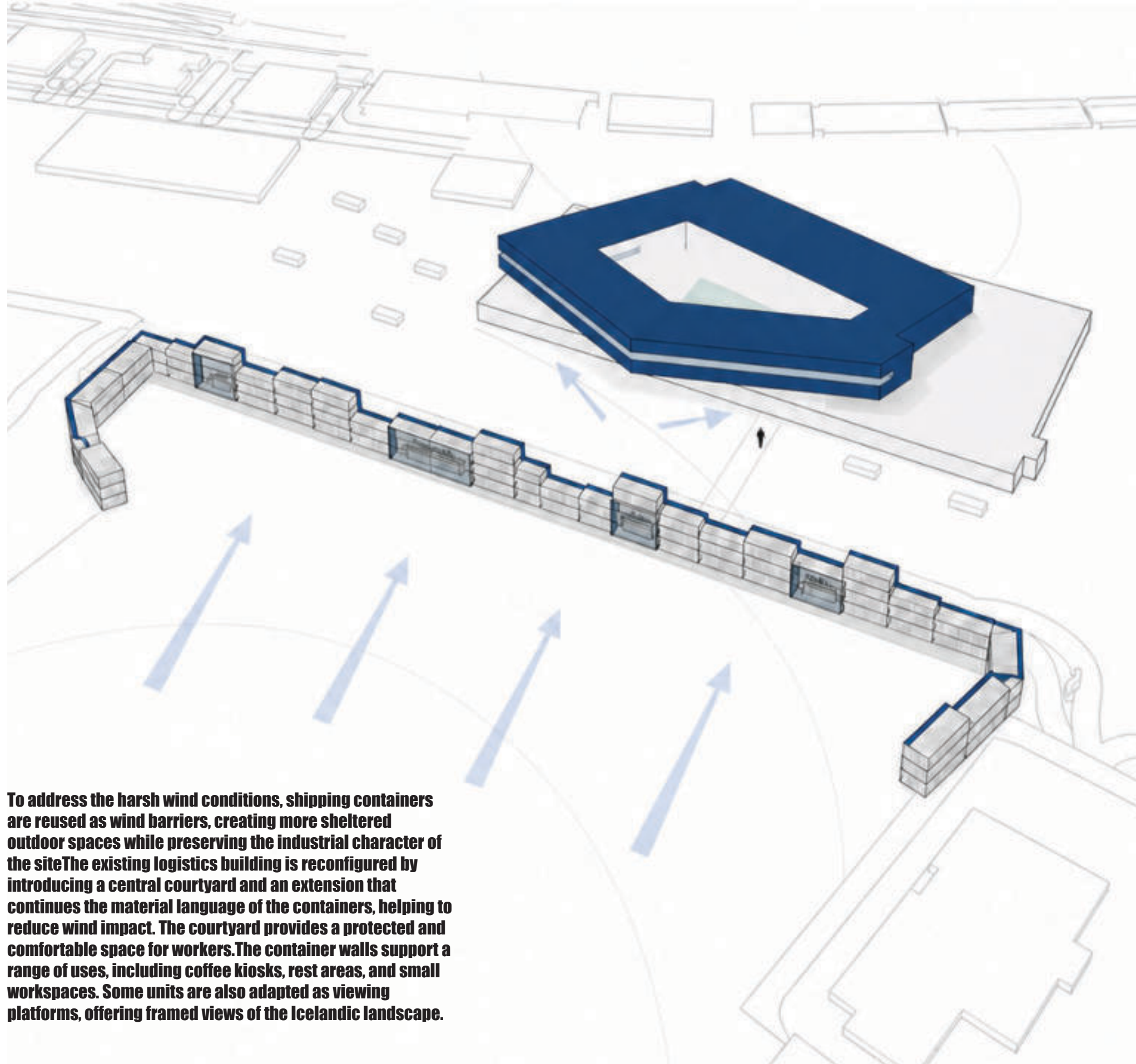
Wind rose analysis shows stronger and more directional winds during winter, primarily from the eastern sector. This informed subsequent wind simulations and the development of a container-based wind buffer.



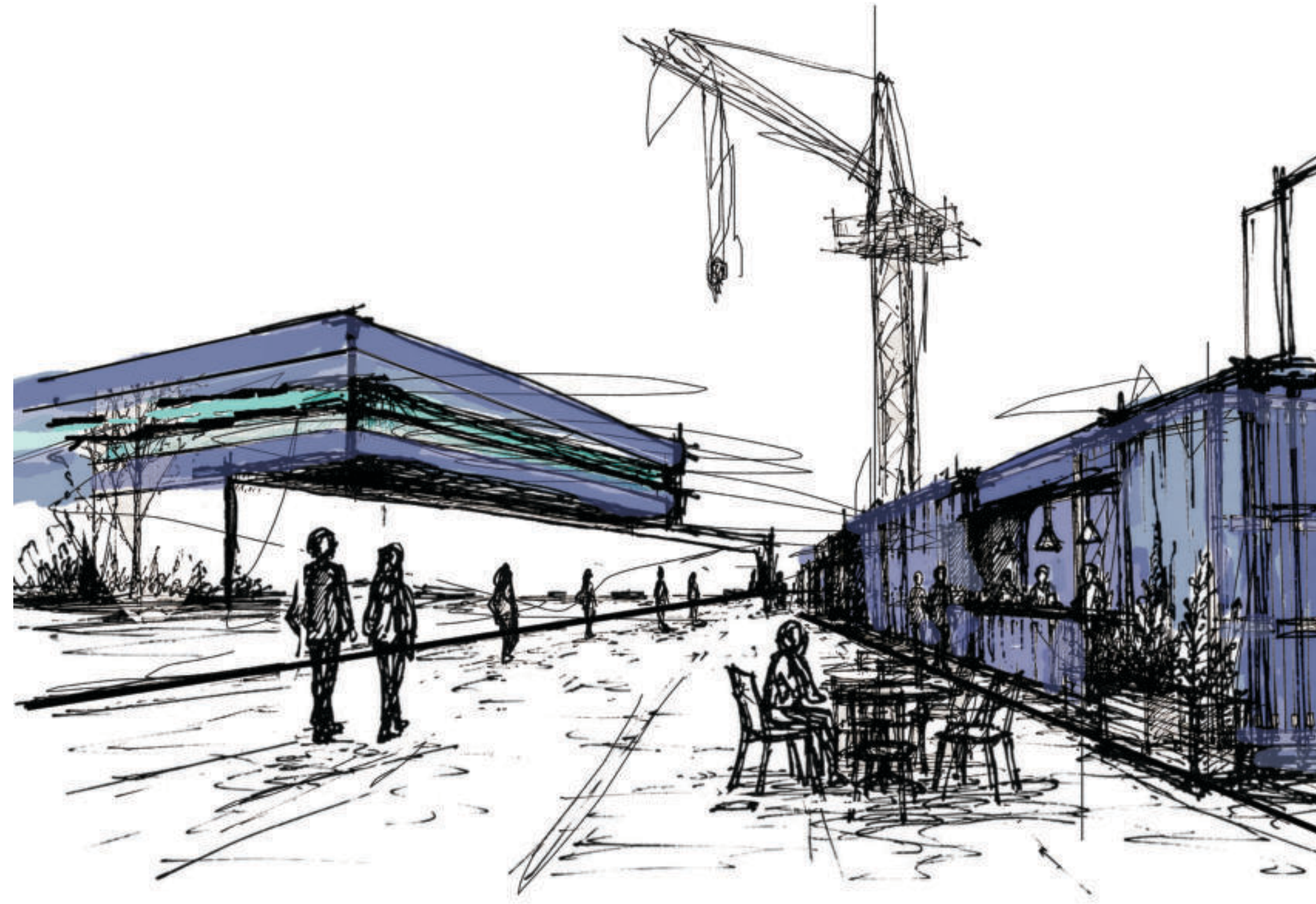
A step-by-step transformation showing how repurposed containers are used to form a wind buffer, converting an exposed industrial site into a sheltered and usable public space.

Wind test Ladybug : // Typology Diagram

Wind Rós

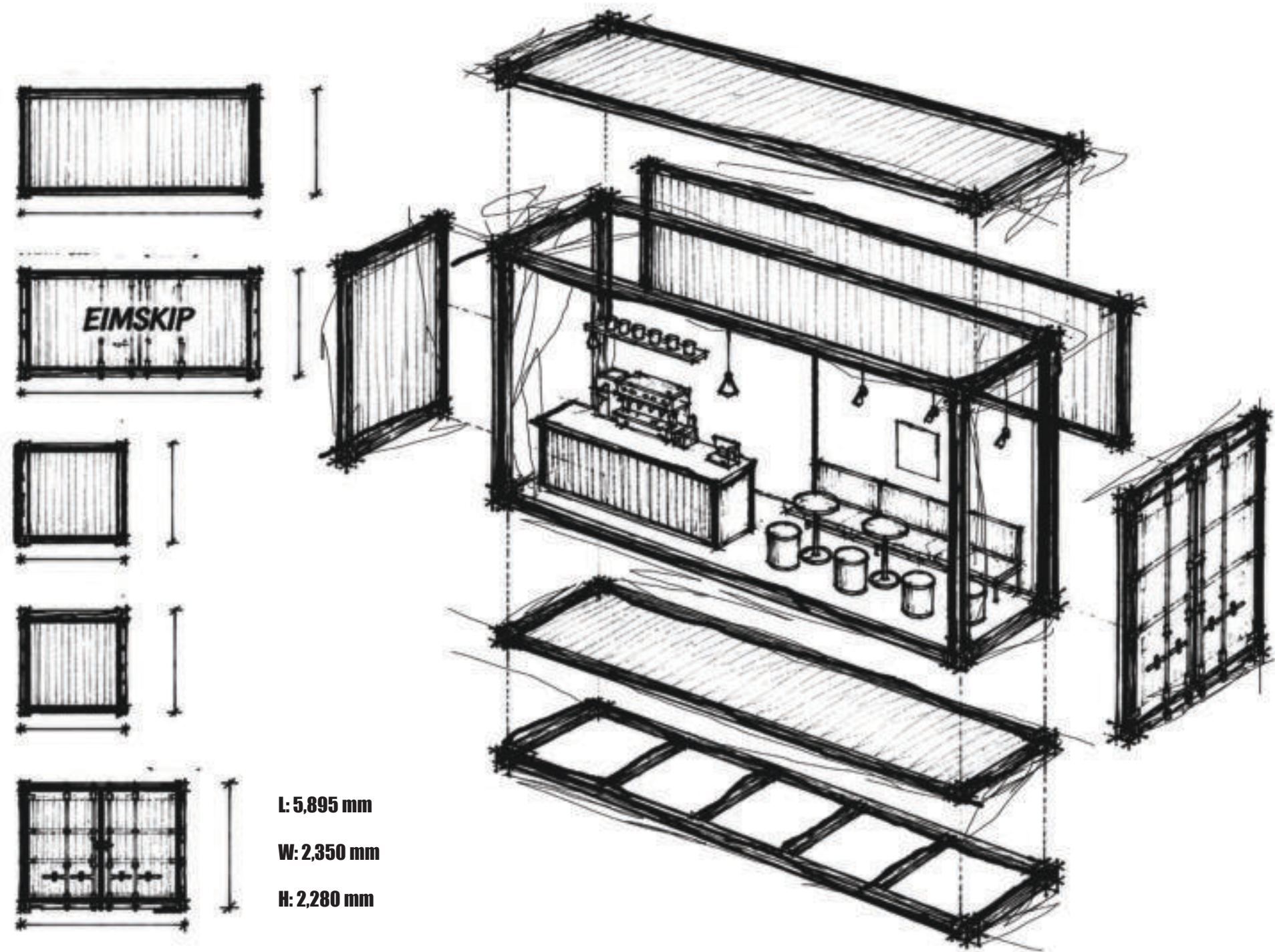


To address the harsh wind conditions, shipping containers are reused as wind barriers, creating more sheltered outdoor spaces while preserving the industrial character of the site. The existing logistics building is reconfigured by introducing a central courtyard and an extension that continues the material language of the containers, helping to reduce wind impact. The courtyard provides a protected and comfortable space for workers. The container walls support a range of uses, including coffee kiosks, rest areas, and small workspaces. Some units are also adapted as viewing platforms, offering framed views of the Icelandic landscape.

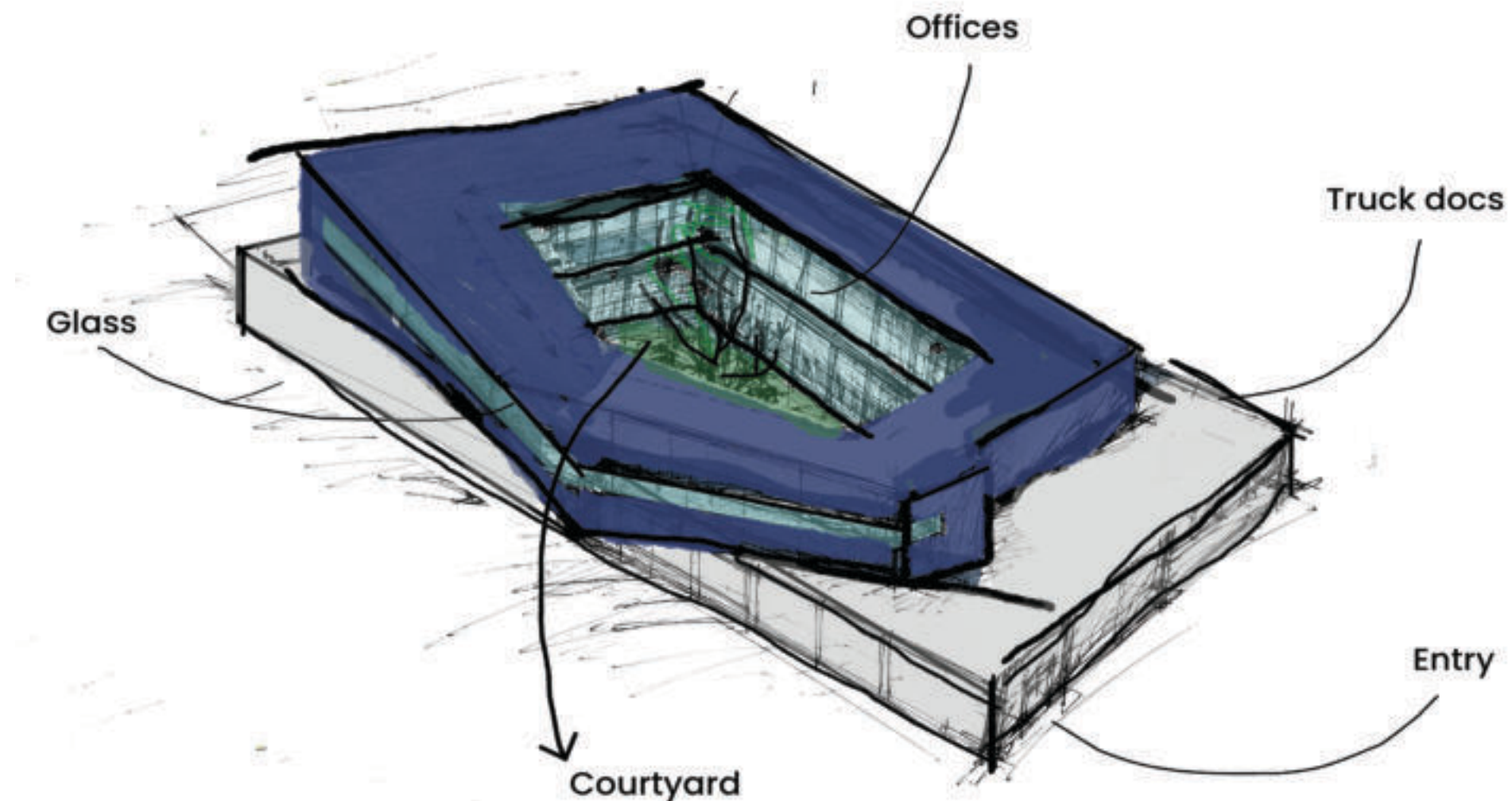


Axonometric Diagram// Perspective sketch

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Exploded axonometric of a reused shipping container adapted into a coffee unit.

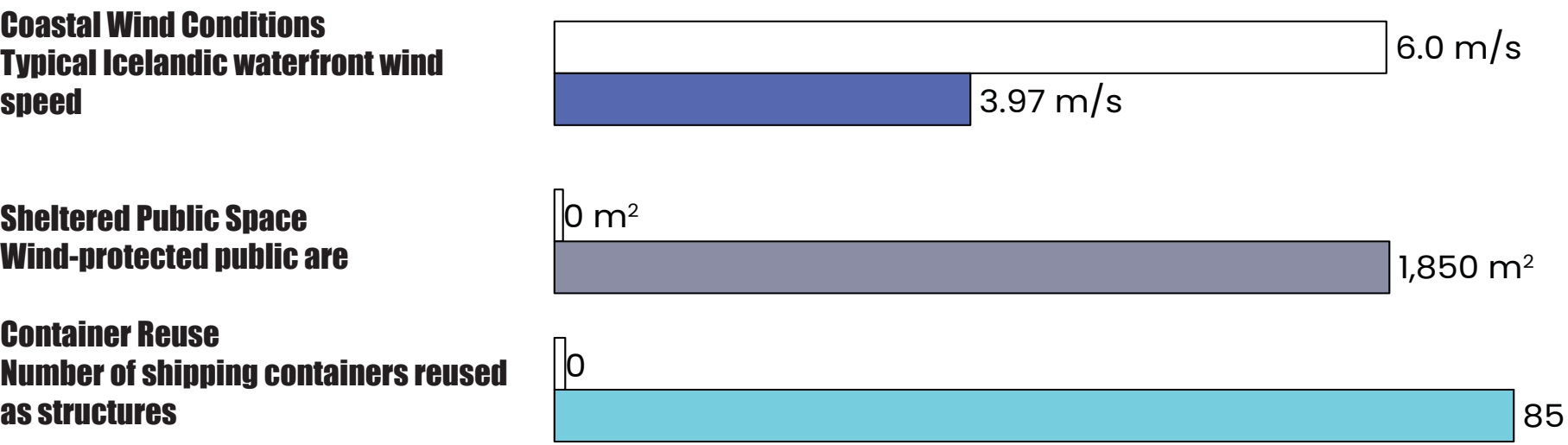


The building organizes its program along the perimeter, enclosing a protected courtyard for workers. The courtyard is designed as a social space, with seating made from recycled container materials.



Exploded View// Program

Wind Rós



Throughout the semester, a series of KIPs were developed. The most relevant strategies were integrated into the project, contributing to a positive outcome.



KPI// Visualization

